INCH-POUND

A-A-59274B
13 September 2005
SUPERSEDES
A-A-59274A
7 December 2004

COMMERCIAL ITEM DESCRIPTION
GENERATOR, SIGNAL
9 kHz to 3.2 GHz

1. SCOPE. This Commercial Item Description (CID) describes a 9 kHz to 3.2 GHz Signal Generator with AM, FM, pulse modulation (PM) and sweep capabilities. This CID is meant as a minimum requirement for the signal generator in which only those manufacturers that meet or surpass the following requirements are qualified per this CID.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation and product sample testing, if so requested, within the accuracies, limits, and specifications herein.

2.1 Classification. Equipment covered by this PD may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be Class 4 in accordance with MIL-PRF-28800, except as specified herein.

2.2 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

2.2.1 Temperature Operating. The equipment shall conform to its specified performance and accuracy while being operated as a volt meter in a temperature range of 0 °C to 40 °C.

2.2.2 Temperature Non-Operating. The equipment shall conform to its specified performance and accuracy after being stored in an environment in the temperature range of -40 °C to 71 °C.

Comments, suggestions, or questions on this document should be addressed to: 542 ATSG/GBTRND 380 Richard Ray Blvd, Ste 104, Robins AFB, GA 31098. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.2.3 Humidity Range Operating. The equipment shall conform to its specified performance and accuracy while being operated in a non-condensing humidity range of 5% to 95% ± 5% RH from 0 °C to 30 °C and 75% ± 5% RH 30 °C to 40 °C and 45% ± 5% RH above 40 °C.

2.3 Electrical Power Sources. The equipment shall operate from nominal single-phase commercial, military, and shipboard power sources of 120 volts AC (±10%) or 240 volts AC (±10%) at line frequencies of 47.5 Hz to 52.5 Hz and 57 Hz to 63 Hz. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein. The equipment shall also have power consumption equal to or less than 400 Volt-Amperes.

2.4 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 4,000 hours of operation with a statistical certainty of 95%.

2.5 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 2000 hours per year.

2.6 Performance Requirements. The instrument shall meet all requirements specified herein, after the warm-up period specified in MIL-PRF-28800, under all combinations of input power conditions, output signal conditions, and operating service condition as specified in MIL-PRF-28800.

2.6.1 Frequency Range. The signal generator shall have a frequency range of 9 kHz to 3.2 GHz with a resolution of 0.1 Hz.

2.6.2 Accuracy. The accuracy of this instrument shall be $< 1 \times 10^{-7}$ /year.

2.6.3 Switching Speed. The switching speed of this instrument shall be as follows:

<1001 MHz: <75 ms
=1001 MHz: <100 ms

2.6.4 Internal Reference Oscillator Accuracy and Stability. The instrument shall contain an internal reference oscillator with accuracy and stability of:

Aging: < ± 0.1 ppm/yr
Temperature: < ± 0.01 ppm

2.6.5 Output Range. The output range on this instrument shall be +10 dBm to -136 dBm.

2.6.6 Display Resolution. The display resolution on this instrument shall be 0.1 dB.

2.6.7 Output Impedance. The output impedance shall be 50 Ohms.

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2.6.8 Reverse Power Protection. The reverse power protection shall be as follows:

=2000 MHz: 50 watts
=3200 MHz: 25 watts

2.6.9 Accuracy. The instrument shall have an output accuracy as follows:

=2500 MHz: ± 1.0 dB
=3200 MHz: ± 1.5 dB

2.6.10 Harmonics. The instrument shall have < -30 dBc harmonics with output [+4 dBm.

2.6.11 Subharmonics. The instrument shall have subharmonics as follows with output = +4 dBm:

<1001 MHz : <-60 dBc
≤3200 MHz : <-50 dBc

2.6.12 Nonharmonics. The instrument shall have nonharmonic levels as follows at = 5 kHz and output = +4 dBm:

< 249 MHz : <-55 dBc
< 1001 MHz : <-60 dBc
< 2001 MHz : <-54 dBc

2.6.13 Residual FM. The instruments shall have residual FM as follows:

< 249 MHz: < 7 Hz
< 501 MHz: < 4 Hz
< 1001 MHz: < 7 Hz
< 2001 MHz: < 14 Hz

2.6.14 Single Side Band (SSB) Phase noise. The instrument shall have SSB phase noise at 20 kHz offset as follows:

500 MHz : <- 120 dBc/ Hz
1000 MHz : <- 116 dBc/ Hz
2000 MHz : <- 110dBc/ Hz
3000 MHz : <- 106 dBc/ Hz

2.6.15 Amplitude Modulation (AM) Range. The amplitude modulation range of this instrument shall be 0 to 99.9% (output [+4 dBm).

2.6.16 AM Resolution. The amplitude modulation resolution of this instrument shall be 0.1%.

2.6.17 AM Accuracy. The amplitude modulation accuracy at 1 kHz shall be ± 5% of setting ± 1.5%.

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2.6.18 AM Rates. The amplitude modulation rates shall be as follows:

Internal: 400 Hz or 1 kHz
External: DC: dc to 25 kHz (typical, 3 dB BW)
AC: 1 Hz to 25 kHz (typical, 3 dB BW)

2.6.19 AM Distortion. The amplitude modulation distortion shall be as follows: (1kHz rate, THD + N, 0.3 to 3 kHz BW)

30% AM : < 2%
70% AM : <3%

2.6.20 Frequency Modulation (FM) Peak Deviation. The frequency modulation peak deviation shall be a follows: (rates > 25 Hz ac FM)

> 500 MHz: 0 to 100 kHz
< 1001 MHz: 0 to 200 kHz
< 2001 MHz: 0 to 400 kHz
≤ 3200 MHz: 0 to 800 kHz

2.6.21 FM Resolution. The frequency modulation resolution shall be as follows:

For = 10% peak deviation
< 2001 MHz: 10 Hz
≥ 2001 MHz: 20 Hz

For > 10% peak deviation
< 2001 MHz: 100 Hz
≥ 2001 MHz: 200 Hz

2.6.22 FM Deviation Accuracy. The frequency modulation deviation accuracy shall be as follows: (internal 1 kHz rate)

< 1001 MHz: ±4% of FM deviation ±30 Hz
< 2001 MHz: ±4% of FM deviation ±60 Hz
≤ 3200 MHz: ±4% of FM deviation ±120 Hz

2.6.23 FM Rates. The frequency modulation rates shall be as follows:

Internal: 400 Hz or 1 kHz
External: DC: dc to 150 kHz (typical, 3 dB BW)
AC: 1 Hz to 150 kHz (typical, 3 dB BW)

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2.6.24 FM Distortion. The frequency modulation distortion shall be as follows: (1 kHz rate, THD + N, 0.3 to 3 kHz BW)

< 1001 MHz: < 1% at deviations > 4 kHz
< 2001 MHz: < 1% at deviations > 8 kHz
≤ 3200 MHz: < 1% at deviations > 16 kHz

2.6.25 Phase Modulation (ϕM) Peak Deviation. The phase modulation peak deviation shall be as follows:

> 500 MHz: 0 to 5 radians
< 1001 MHz: 0 to 10 radians
< 2001 MHz: 0 to 20 radians
≤ 3200 MHz: 0 to 40 radians

2.6.26 ϕM Resolution. The phase modulation resolution shall be as follows:

< 2001 MHz: 0.01 radians
≥ 2001 MHz: 0.02 radians

2.6.27 ϕM Deviation Accuracy. The phase modulation deviation accuracy shall be as follows: (internal 1 kHz rate)

< 1001 MHz: ±4% of deviation ±0.05 radians
< 2001 MHz: ±4% of deviation ±0.1 radians
≤ 3200 MHz: ±4% of deviation ±0.2 radians

2.6.28 ϕM Rates. The phase modulation rates shall be as follows:

Internal: 400 Hz or 1 kHz
External: 20 Hz to 10 kHz (typical, 3 dB BW)

2.6.29 ϕM Distortion. The phase modulation distortion shall be as follows: (1 kHz rate)

< 1001 MHz: < 4% at deviations = 3 radians
< 2001 MHz: < 4% at deviations = 6 radians
≤ 3200 MHz: < 4% at deviations = 12 radians

2.6.30 Internal Modulation Source. The internal modulation source shall be 400 Hz, front panel BNC connector provided at nominally 1 Vpk into 600 Ohms.

2.6.31 External Modulation Source. The external modulation source shall be 1 Vpk into 600 Ohms (nominal) required for full scale modulation.

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2.6.32 Pulse Modulation (PM) On/Off Ratio. The pulse modulation On/Off ratio shall be as follows:

< 2000 MHz: > 80 dB
= 3200 MHz: > 70 dB

2.6.33 PM Rise/Fall Times. The pulse modulation rise/fall time shall be < 20 ns.

2.6.34 PM Repetition Rate. The maximum pulse modulation repetition rate shall be 2.5 MHz.

2.6.35 PM Delay. The pulse modulation delay shall be less than 60 ns.

2.6.36 PM Input. The pulse modulation input shall be TTL level (± 15V max).

2.6.37 Remote Operation. Equipment shall interface with GPIB (IEEE-488.2) Listen and Talk.

2.6.38 Storage Registers. Equipment shall have 300 storage registers with sequence and register number displayed.

2.6.39 Input Connections. The instrument shall provide Type-N connectors and five-way binding posts. The connectors shall be able to cover all operational modes of the instrument.

2.6.40 Dimensions. The width and depth of the equipment shall be compatible with mounting in an ANSI/EIA-310-D standard equipment rack. The height shall be compatible with four standard height units, 4U. One unit or "U" is defined by ANSI/EIA-310-D to be 1.75 inches or 44.45 mm.

2.6.41 Weight. The total weight of the equipment, excluding accessories and manuals, shall not exceed 19 pounds, (8.5 kilograms).

2.6.42 Rack Mount. The analyzer shall be capable of being rack-mounted in a standard ANSI/EIA-310-D equipment rack and include a rack-mount conversion kit if needed.

2.7 Manuals. The equipment shall be delivered with operator, maintenance, calibration, and illustrated parts manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this PD, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

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4.2 The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING.

5.1 Preservation, Packing, and Marking. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

6.2 Sources of documents.

6.2.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained at http://assist.daps.dla.mil/ or from the Standardization Documents Order Desk, 700 Robbins Ave., Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.2.1.1 MIL-PRF-28800F, 24 June 1996, Test Equipment for Use with Electrical and Electronic Equipment, General Specification for

6.2.2 The Code of Federal Regulations, (CFR), may be at http://www.gpoaccess.gov/cfr/index.html or obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.2.3 Electronics Industry Alliance (EIA). Hardcopy of EIA standards and technical publications are available from Global Engineering Documents, the exclusive distributor of EIA standards. Web site: http://global.ihs.com Phone number: 1-800-854-7179.

ANSI/EIA-310-D, 24 August 1992, Cabinets, Racks, Panels, and Associated Equipment

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MILITARY INTEREST

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 84

Agent:
Air Force - 99

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